

The Evolution of Artificial Intelligence and Deep Learning

In 2014, DeepMind Technologies, an artificial intelligence company co-founded by Demis Hassabis, Shane Legg, and Mustafa Suleyman in London, was acquired by Google for over 00 million. Under Google, DeepMind developed AlphaGo, an AI system that conquered the game of Go in 2016 by defeating world champion Lee Sedol in Seoul, South Korea.

Meanwhile, in 2015, OpenAI was founded in San Francisco by Sam Altman, Elon Musk, Greg Brockman, Ilya Sutskever, and Wojciech Zaremba as a non-profit AI research organization. OpenAI introduced Generative Pre-trained Transformers (GPT), revolutionizing natural language processing.

Transformer Architecture and Attention Mechanisms

In 2017, a team of researchers at Google Brain and Google Research, including Ashish Vaswani, Noam Shazeer, Niki Parmar, and Jakob Uszkoreit, published the landmark paper *Attention Is All You Need*. The paper introduced the Transformer architecture, replacing recurrent neural networks (RNNs) and convolutional neural networks (CNNs) in sequence modeling tasks.

In 2023, Google integrated Google Brain and DeepMind into Google DeepMind, led by CEO Demis Hassabis. Google DeepMind developed the Gemini series of multimodal large language models to compete with GPT-4, which was developed by OpenAI and partnered with Microsoft. Microsoft invested billions of dollars into OpenAI and integrated GPT models into Microsoft Copilot.